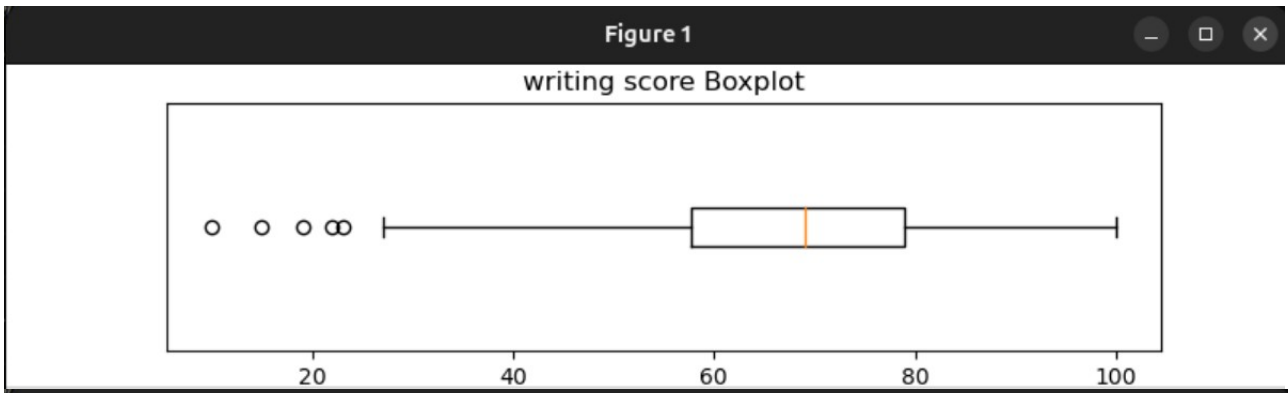
Figure 1
writing score Boxplot
20
40
60
80
100

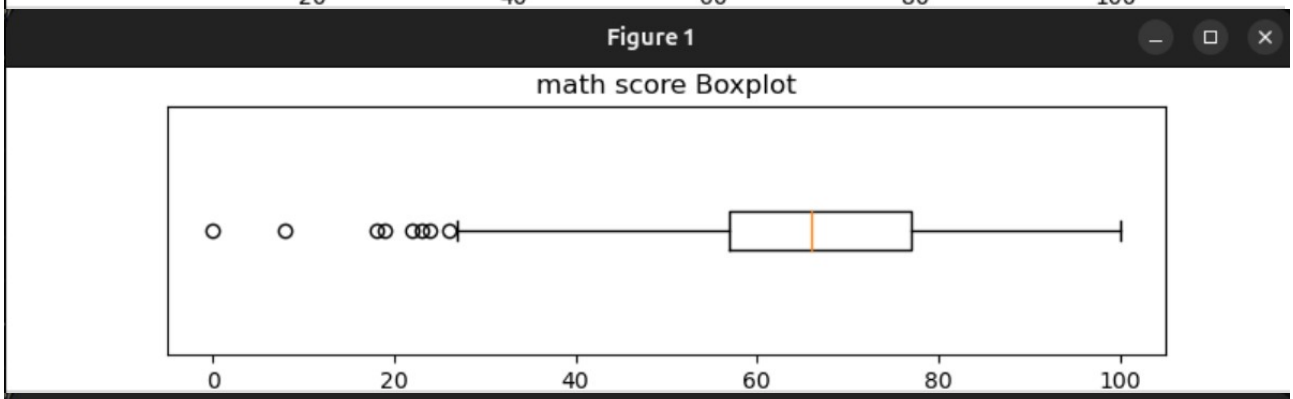
Figure 1
math score Boxplot
0
20
40
60
80
100

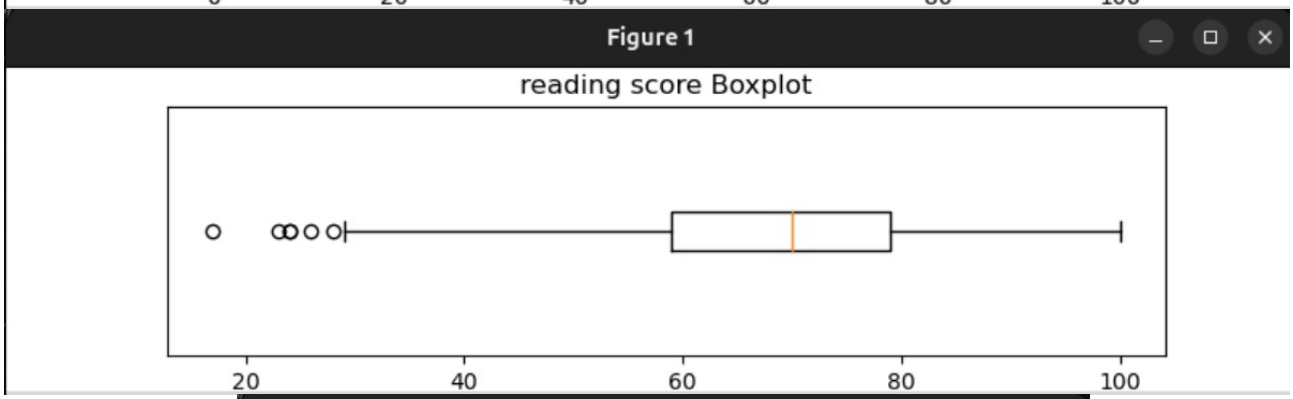
Figure 1
reading score Boxplot
20
40
60
80
100

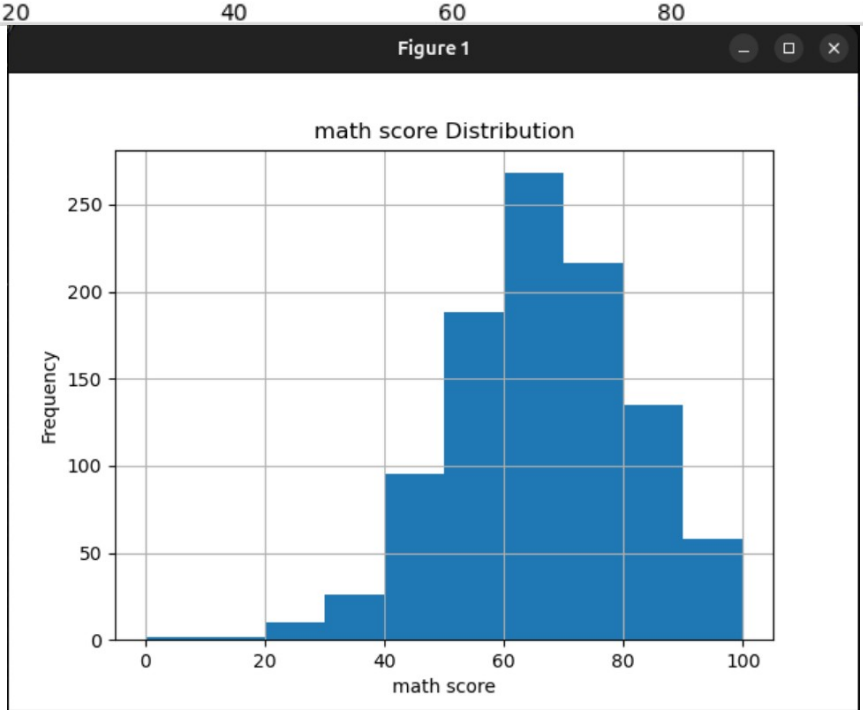
Figure 1
math score Distribution
Frequency
0
50
100
150
200
250
0
20
40
60
80
100
math score

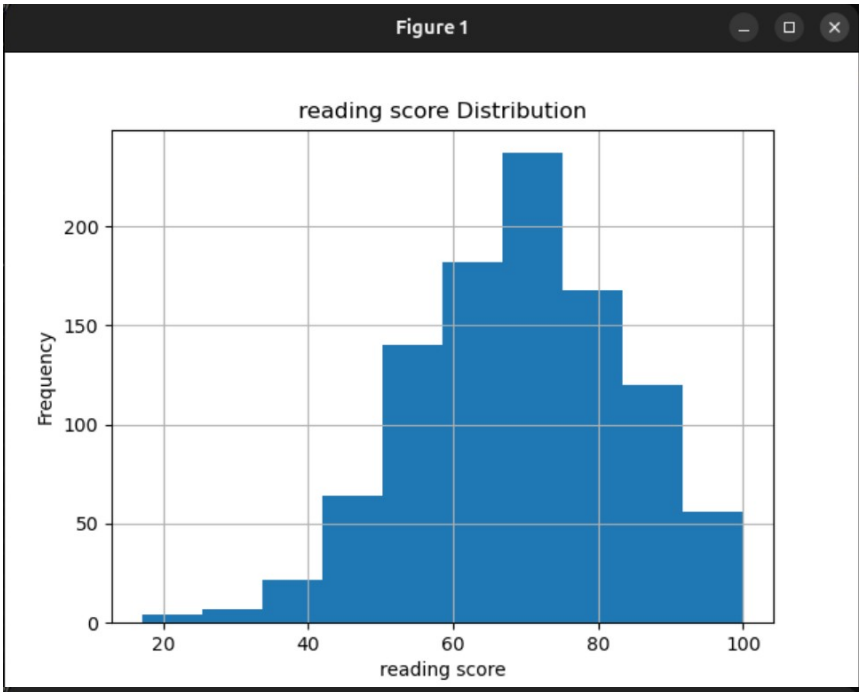
Figure 1
reading score Distribution
Frequency
0
50
100
150
200
20
40
60
80
100
reading score

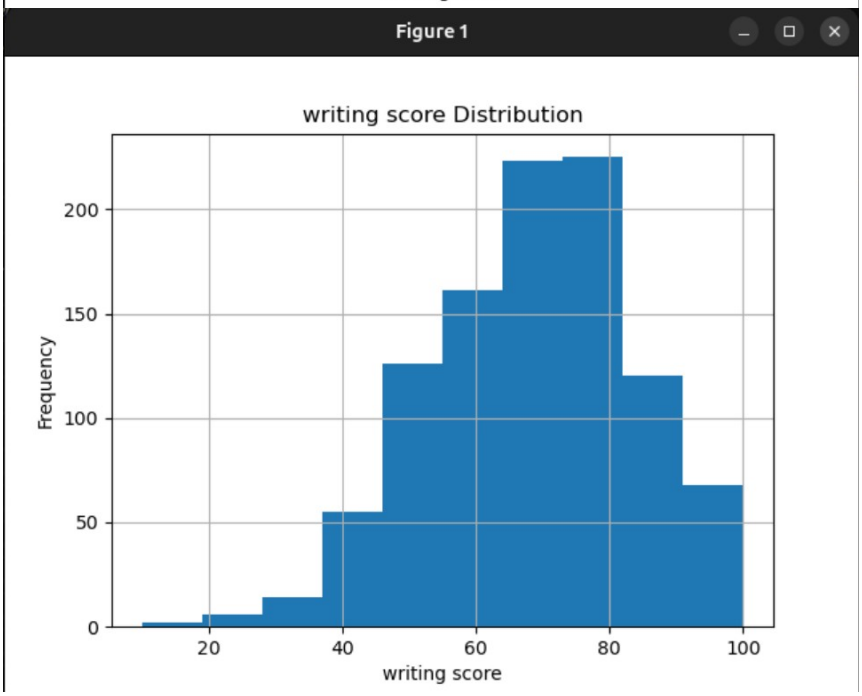
Figure 1
writing score Distribution
Frequency
0
50
100
150
200
20
40
60
80
100
writing score

```
[Running] python -u "/home/swaraj/Documents/import pandas as pd.py"
gender race/ethnicity parental level of education        lunch test preparation course  math score  reading score  writing score
female        group B           bachelor's degree     standard                    none          72             72             74
female        group C                some college     standard               completed          69             90             88
female        group B             master's degree     standard                    none          90             95             93
  male        group A          associate's degree free/reduced                    none          47             57             44
  male        group C                some college     standard                    none          76             78             75
Numerical Columns: ['math score', 'reading score', 'writing score']
Categorical Columns: ['gender', 'race/ethnicity', 'parental level of education', 'lunch', 'test preparation course']

Mean:
 math score       66.089
reading score    69.169
writing score    68.054
dtype: float64

Median:
 math score       66.0
reading score    70.0
writing score    69.0
dtype: float64

Mode:
 math score       65
reading score    72
writing score    74
Name: 0, dtype: int64

Min:
 math score        0
reading score    17
writing score    10
dtype: int64

Max:
 math score       100
reading score    100
writing score    100
dtype: int64

Sum:
 math score       66089
reading score    69169
writing score    68054
dtype: int64

Variance:
 math score       229.918998
reading score    213.165605
writing score    230.907992
dtype: float64

Standard Deviation:
 math score       15.163080
reading score    14.600192
writing score    15.195657
dtype: float64

[Done] exited with code=0 in 0.812 seconds

[Running] python -u "/home/swaraj/Documents/import pandas as pd.py"
Quartiles:
                    math score  reading score  writing score
Q1 (25%)                 57.0           59.0          57.75
Q2 / Median (50%)        66.0           70.0          69.00
Q3 (75%)                 77.0           79.0          79.00

[Done] exited with code=0 in 0.953 seconds

[Running] python -u "/home/swaraj/Documents/import pandas as pd.py"
Correlation Matrix:
                 math score  reading score  writing score
math score           1.00           0.82           0.80
reading score        0.82           1.00           0.95
writing score        0.80           0.95           1.00

Covariance Matrix:
                 math score  reading score  writing score
math score         229.92         181.00         184.94
reading score      181.00         213.17         211.79
writing score      184.94         211.79         230.91
```